Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/33**

Paper 3 Advanced Practical Skills 1 **March 2020**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the March 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5	<u>'List rule' guidance</u> (see examples below) For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided • Any response marked *ignore* in the mark scheme should not count towards ***n*** • Incorrect responses should not be awarded credit but will still count towards ***n*** • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.
6	<u>Calculation specific guidance</u> Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form, (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	<u>Guidance for chemical equations</u> Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Question	Answer	Marks
1(a)	Value of x in range 10.0 to 20.0 cm	**1**
1(b)	Value of T in range 0.50 to 1.50 s	**1**
	Evidence of repeat readings: at least two values of at least $5T$	**1**
1(c)	Six sets of readings of x and T with correct trend and without help scores 4 marks, five sets scores 3 marks etc.	**4**
	Range: $x_{min} \leqslant 12.0$ cm <u>and</u> $x_{max} \geqslant 25.0$ cm	**1**
	Column headings: Each column heading must contain a quantity and a unit where appropriate The presentation of quantity and unit must conform to accepted scientific convention e.g. $1 / x$ (cm^{-1})	**1**
	Consistency: All values of raw x must be given to the nearest mm	**1**
	Significant figures: All values of $1 / x$ should be to the same s.f. as (or one more than) the s.f. in the corresponding x value	**1**
	Calculation: Values of $1 / x$ calculated correctly	**1**

Question	Answer	Marks
1(d)(i)	Axes: Sensible scales must be used, no awkward scales (e.g. 3:10) Scales must be chosen so that the plotted points occupy at least half the graph grid in both *x* and *y* directions Scales must be labelled with the quantity which is being plotted. Scale markings should be no more than 3 large squares apart.	**1**
	Plotting of points: All observations must be plotted on the grid. Diameter of plotted points must be ⩽ half a small square (no blobs). Plots must be accurate to within half a small square in both *x* and *y* directions.	**1**
	Quality: All points in the table must be plotted (at least 5) for this mark to be awarded. Scatter of plots must be no more than $\pm$ 0.5 m^{-1} ($\pm$ 0.005 cm^{-1}) from a straight line in the *x*-direction.	**1**
1(d)(ii)	Line of best fit: Judged by balance of all points on the grid (at least 5) about the candidate's line. There must be an even distribution of points either side of the line along the full length One anomalous point is allowed only if clearly indicated (i.e. circled or labelled) by the candidate. Lines must not be kinked or thicker than half a square.	**1**
1(d)(iii)	Gradient: The hypotenuse of the triangle used must be greater than half the length of the drawn line. Method of calculation must be correct. Both read-offs must be accurate to half a small square in both the *x* and *y* directions.	**1**
	y-intercept: **Either** Correct read-off from a point on the line substituted into $y = mx + c$ or an equivalent expression, with read-off accurate to half a small square in both *x* and *y* directions. **Or** Intercept read directly from the graph, with read-off at x = zero accurate to half a small square in *y* direction.	**1**

Question	Answer	Marks
1(e)	*a* equal to candidate's gradient, and *b* equal to candidate's intercept. *a* and *b* are not fractions. *a* has two or more significant figures.	**1**
1(e)	Units for *a* and *b* correct (e.g. s cm or s m for *a*, and s for *b*).	**1**

Question	Answer	Marks
2(a)	Value of *N*, with no unit, in range 10 to 14.	**1**
2(b)	Value of *L* to nearest mm, with unit, in range 5.0 to 15.0 cm.	**1**
2(c)	Value(s) of raw θ to nearest degree and < 90°.	**1**
	Value of I to nearest mA or nearest 0.1 mA, with unit	**1**
2(d)	Absolute uncertainty in θ value of 2 to 5° and correct method of calculation to obtain percentage uncertainty. If several readings have been taken, then the absolute uncertainty can be half the range (but not zero if values are equal) if the working is clearly shown.	**1**
2(e)	Correct calculation of *B*	**1**
2(f)	Second values of *N* and *L*	**1**
	Second values of θ and I	**1**
	Quality: second θ > first θ	**1**
2(g)(i)	Two values of *k* calculated correctly.	**1**
2(g)(ii)	Justification based on s.f. in θ, I and *L*.	**1**
2(g)(iii)	Sensible comment relating to the calculated values of *k*, testing against a criterion specified by the candidate.	**1**

Question	Answer		Marks
2(h)(i)	Two readings are not enough to draw a valid conclusion Difficult to determine I as reading not steady Parallax error when measuring θ Large % uncertainty in θ Difficult to re-wind and fix the wire / wires may touch / wire is kinked Channel moves on bench	4 max	**4**
2(h)(ii)	Take more readings <u>and</u> plot a graph / calculate more k values and <u>compare</u> <u>Method</u> of cleaning crocodile clips / contacts Method of overcoming parallax error, e.g. measure change of angle by turning channel on bench / measure θ on a photo / angle markings on compass / mark 'before' and 'after' positions <u>on compass</u> Use larger current / voltage Use two ready-wound channels / use new wire Method of fixing channel, e.g. tape channel to bench	4 max	**4**